WORKED SOLUTION

STEP II 2023, Question 4

Pure • Eliminating Surds from Polynomials

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 6.3, *A Polynomial for* $\sqrt{2}+\sqrt{3}+\sqrt{5}$: https://beyondstep.keithmcnulty.org/ch06/field-extensions.html

The key idea

To clear a surd from an equation, put everything containing the surd on one side and everything else on the other, then square (or cube). Each part is the same move. Write the number as "something nice plus a surd", isolate that surd, and raise to a power until it disappears. When a surd survives after expanding, collect its terms on one side and repeat.

Solution

Part (i)

Expanding $(x-\sqrt{2})^2 = 3$ gives $x^2 - 2\sqrt{2}\,x + 2 = 3$, that is

$$x^2 - 1 = 2\sqrt{2}\,x.$$

The surd is now on its own, so we square both sides:

$$x^4 - 2x^2 + 1 = 8x^2 \quad \Longrightarrow \quad x^4 - 10x^2 + 1 = 0,$$

as required. (Squaring only goes one way, which is all we need: every x satisfying the first equation satisfies the second.)

For the deduction, take $x = \sqrt{2}+\sqrt{3}$. Then $x - \sqrt{2} = \sqrt{3}$, so $(x-\sqrt{2})^2 = 3$, and by what we have just shown $\mathrm{f}(\sqrt{2}+\sqrt{3}) = 0$.

Part (ii)

Let $x = \sqrt{2}+\sqrt{3}+\sqrt{5}$ and $y = x - \sqrt{5} = \sqrt{2}+\sqrt{3}$. By part (i), $y^4 - 10y^2 + 1 = 0$. We now write this in terms of x. First,

$$y^2 = (x-\sqrt{5})^2 = (x^2+5) - 2\sqrt{5}\,x,$$

and so

$$\begin{aligned} y^4 &= (x^2+5)^2 - 4\sqrt{5}\,x(x^2+5) + 20x^2 \\ &= x^4 + 30x^2 + 25 - 4\sqrt{5}\,x^3 - 20\sqrt{5}\,x. \end{aligned}$$

Therefore

$$\begin{aligned} 0 &= y^4 - 10y^2 + 1 \\ &= \left(x^4 + 30x^2 + 25 - 4\sqrt{5}\,x^3 - 20\sqrt{5}\,x\right) - 10\left(x^2 + 5 - 2\sqrt{5}\,x\right) + 1 \\ &= x^4 + 20x^2 - 24 - 4\sqrt{5}\,x^3. \end{aligned}$$

The $\sqrt{5}\,x$ terms cancel, which is lucky. Only one surd term remains, so we isolate it and square:

$$x^4 + 20x^2 - 24 = 4\sqrt{5}\,x^3 \quad\Longrightarrow\quad (x^4 + 20x^2 - 24)^2 = 80x^6.$$

Expanding the left-hand side,

$$\begin{aligned} (x^4 + 20x^2 - 24)^2 &= x^8 + 40x^6 + 400x^4 - 48x^4 - 960x^2 + 576 \\ &= x^8 + 40x^6 + 352x^4 - 960x^2 + 576. \end{aligned}$$

Subtracting $80x^6$, we can take

$$g(x) = x^8 - 40x^6 + 352x^4 - 960x^2 + 576,$$

which has degree 8 and integer coefficients, with $g(\sqrt{2}+\sqrt{3}+\sqrt{5}) = 0$.

Ask yourself Why do we eliminate $\sqrt{5}$ last, using part (i) for $\sqrt{2}+\sqrt{3}$, rather than starting from scratch?

Part (i) has already removed two of the three surds, so we only have to deal with one more. Removing a single surd by squaring doubles the degree, from 4 to 8, which is exactly the degree we were asked for. Starting from scratch would mean juggling all three surds at once.

Part (iii)

Let $x = a + \sqrt{2}$, so $a = x - \sqrt{2}$, and $a^3 - 3a + 1 = 0$. Expanding,

$$(x-\sqrt{2})^3 = x^3 - 3\sqrt{2}\,x^2 + 6x - 2\sqrt{2},$$

so

$$0 = (x-\sqrt{2})^3 - 3(x-\sqrt{2}) + 1 = x^3 + 3x + 1 - \sqrt{2}\,(3x^2 - 1).$$

We isolate the surd and square:

$$x^3 + 3x + 1 = \sqrt{2}\,(3x^2 - 1) \quad \Longrightarrow \quad (x^3 + 3x + 1)^2 = 2(3x^2 - 1)^2.$$

The two sides are

$$(x^3 + 3x + 1)^2 = x^6 + 6x^4 + 2x^3 + 9x^2 + 6x + 1,$$
$$2(3x^2 - 1)^2 = 18x^4 - 12x^2 + 2,$$

so we can take

$$\mathrm{h}(x) = x^6 - 12x^4 + 2x^3 + 21x^2 + 6x - 1.$$

We used only the fact that a satisfies $t^3 - 3t + 1 = 0$, so the same working applies to b and to c. Hence $\mathrm{h}(a+\sqrt{2}) = \mathrm{h}(b+\sqrt{2}) = \mathrm{h}(c+\sqrt{2}) = 0$.

Part (iv)

Squaring will not remove cube roots, so we need a different identity. Let $p = \sqrt[3]{2}$, $q = \sqrt[3]{3}$ and $x = p + q$. The binomial expansion gives

$$x^3 = p^3 + q^3 + 3pq(p+q) = 5 + 3pq\,x.$$

Ask yourself How do we get rid of pq, which is still irrational?

Here $pq = \sqrt[3]{6}$ is the only irrational quantity left, and it appears only as a factor. We isolate it and cube, using $(pq)^3 = 6$:

$$x^3 - 5 = 3pq\,x \quad \Longrightarrow \quad (x^3 - 5)^3 = 27 \cdot 6\,x^3 = 162x^3.$$

Since $(x^3 - 5)^3 = x^9 - 15x^6 + 75x^3 - 125$, we can take

$$\mathrm{k}(x) = x^9 - 15x^6 - 87x^3 - 125,$$

which has integer coefficients, and $\mathrm{k}(\sqrt[3]{2} + \sqrt[3]{3}) = 0$.

Checks

- In part (ii), $\sqrt{2} + \sqrt{3} + \sqrt{5} \approx 5.38233$. With $x^2 \approx 28.9695$, we get $x^8 \approx 704\,310$ and $40x^6 \approx 972\,485$. Then $352x^4 \approx 295\,410$ and $960x^2 \approx 27\,811$, and the terms of g cancel to within rounding error.
- Changing the signs of $\sqrt{2}$, $\sqrt{3}$ and $\sqrt{5}$ never changes $\mathrm{g}(x)$, so all eight numbers $\pm\sqrt{2} \pm \sqrt{3} \pm \sqrt{5}$ are roots. That accounts for the degree 8. Similarly, h has the six roots $a \pm \sqrt{2}$, $b \pm \sqrt{2}$, $c \pm \sqrt{2}$.
- The roots of $t^3 - 3t + 1 = 0$ are $2\cos 40^\circ$, $2\cos 160^\circ$ and $2\cos 280^\circ$. For example, $a \approx 1.53209$ gives $a + \sqrt{2} \approx 2.94630$, and $\mathrm{h}(2.94630)$ is zero to within rounding error.
- In part (iv), $\sqrt[3]{2} + \sqrt[3]{3} \approx 2.70217$, and $\mathrm{k}(2.70217)$ is zero to within rounding error. A computer check confirmed all four polynomials to about 10^{-10}.

Common slips

- Squaring before isolating the surd. If $\sqrt{5}$ terms are left on both sides, they will not cancel.
- In part (iv), trying to square, or cubing x without spotting that $x^3 = 5 + 3pqx$. The expansion $(p+q)^3 = p^3 + q^3 + 3pq(p+q)$ is the whole trick.
- Sign errors in the cubic expansion in part (iii), especially in the constant $-2\sqrt{2} + 3\sqrt{2} = \sqrt{2}$.

Beyond STEP

A number that is a root of a polynomial with integer coefficients is called *algebraic*. This question shows by hand that sums of algebraic numbers are algebraic. In general, the proof uses linear algebra: $\mathbb{Q}(\alpha, \beta)$ is a finite-dimensional vector space over $\mathbb{Q}$. So the powers $1, \alpha+\beta, (\alpha+\beta)^2, \ldots$ must eventually be linearly dependent. The degrees 4, 8, 6 and 9 that appear here are the dimensions of these *field extensions*. For more on this, see Allenby, *Rings, Fields and Groups*.

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